Combinatorial Processes & Invariants

Kevin Hu

Part 2 Combinatorics · Invariants · Games

In a process problem a board, a sequence or a set of numbers changes by fixed rules, and you have to decide what can happen. Section 1 describes the two approaches, construction and state functions (invariants, monovariants and colourings). Section 2 has nine problems, from the MAT to RMM 2021, and Section 3 has their solutions. Problems of this kind also come up in Oxford and Cambridge interviews for Mathematics and Computer Science.
Prerequisites. Induction, parity and modular arithmetic.

1 Combinatorial processes

A process problem has the following shape.

$$\text{Initial state} + \text{Operation rules} \longrightarrow \text{Final state}$$

The number of possible configurations grows very quickly, so checking every case is hopeless except for the smallest boards. There are two approaches.

1.1 The constructive approach

When the question asks you to reach a given state, build the sequence of moves. Three things help.

1. Special cases. Work out small systems ($n = 1, 2, 3$) by hand and look for a pattern.
2. Induction. Show that if a state is reachable for n, then the corresponding state is reachable for $n + 1$.
3. Combined operations. Find a macro-operation, a short sequence of basic moves whose total effect is a simple, predictable change.

1.2 State functions

When the question asks you to prove that a state cannot be reached, or that a process stops, map each state to a single quantity f, the *state function*, and follow f instead of the whole board. Most state functions come from one of the following.

1. Colourings, for example a chessboard colouring and the resulting parity.
2. Weights or valuations, for example assigning prime values or algebraic terms to positions and adding them up.

We want a state function f on which every valid operation has a simple, predictable effect. There are two kinds.

© 2026 Kevin Hu · kevinvariant08.github.io · CC BY-NC-ND 4.0

1. Invariants. The value of f stays the same under every operation.
2. Monovariants, also called semi-invariants. Every operation moves f in the same direction, so f never increases or f never decreases. These are the usual way to prove that a process must terminate.

Choosing f is a matter of balance between two extremes.

The state itself	**Constant function**
Contains *all* information. Far too many cases.	Contains *no* information. Too few cases to be useful.

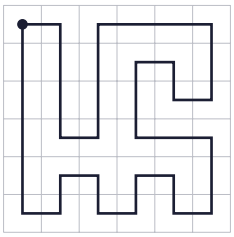
A useful invariant lies in the middle. The invariant forgets most of the full state but keeps enough structure to show that the target configuration cannot be reached.

2 Problems

Problem 1 (MAT 2009) Given an $n \times n$ grid of squares, where $n > 1$, a *tour* is a path drawn within the grid such that

- along its way the path moves, horizontally or vertically, from the centre of one square to the centre of an adjacent square,
- the path starts and finishes in the same square,
- the path visits the centre of every other square just once.

For example, below is a tour drawn in a 6×6 grid of squares which starts and finishes in the top-left square.



For parts (i) to (iv) it is assumed that n is *even*.

(i) With the aid of a diagram, show how a tour, which starts and finishes in the top-left square, can be drawn in any $n \times n$ grid.

(ii) Is a tour still possible if the start/finish point is changed to the centre of a different square? Justify your answer.

Suppose now that a robot is programmed to move along a tour of an $n \times n$ grid. The robot understands two commands,

- command R which turns the robot clockwise through a right angle,
- command F which moves the robot forward to the centre of the next square.

The robot has a program, a list of commands, which it performs in the given order to complete a tour. Say that, in total, command R appears r times in the program and command F appears f times.

© 2026 Kevin Hu · kevinvariant08.github.io · CC BY-NC-ND 4.0

(iii) Initially the robot is in the top-left square pointing to the right. Assuming the first command is an F, what is the value of f? Explain also why $r+1$ is a multiple of 4.

(iv) Must the results of part (iii) still hold if the robot starts and finishes at the centre of a different square? Explain your reasoning.

(v) Show that a tour of an $n \times n$ grid is not possible when n is odd.

Problem 2 (MAT 2020 extension) During a holiday consisting of exactly n sunny days and n rainy days, Miriam and Adam eat sweets according to the following rules.

- On any given day, Miriam eats a number of sweets equal to the total number of sunny days that have occurred so far during the holiday (including the current day).
- Adam eats sweets only on rainy days. If the kth day of the holiday is rainy, he eats exactly k sweets.

Prove that by the end of the holiday, Miriam and Adam will have eaten the same total number of sweets, regardless of the sequence of the weather.

Problem 3 (BMO1 2024) Rhian and Jack are playing a game in which initially the number 10^6 is written on a blackboard. If the current number on the board is n, a move consists of choosing two different positive integers a, b such that $n = ab$ and replacing n with $|a-b|$. Rhian starts, then the players make moves alternately. A player loses if they are unable to move. Determine, with proof, which player has a winning strategy.

Problem 4 Write the numbers $1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{2018}$ on a blackboard. Each time, you choose two different numbers x and y, erase them, and write a new number $x+y+xy$. This operation is repeated until exactly one number remains on the board. Prove that the final remaining number is independent of the order of operations, and find the exact value of this number.

Problem 5 (RMM 2021) Consider an integer $n \geq 2$ and write the numbers $1, 2, \ldots, n$ down on a board. A move consists in erasing any two numbers a and b, and, for each $c \in \{a+b, |a-b|\}$, writing c down on the board, unless c is already there. If c is already on the board, do nothing. For all integers $n \geq 2$, determine whether it is possible to be left with exactly two numbers on the board after a finite number of moves.

Problem 6 (Russia 2022) Initially, the pair of numbers $(1, 1)$ is written on a blackboard. For any two real numbers x and y, if one of the pairs $(x, y-1)$ or $(x+y, y+1)$ is already written on the board, you may write the other pair. Similarly, if one of the pairs (x, xy) or $(1/x, y)$ is already on the board, you may write the other pair. Prove that the first number in every pair written on the board must always be positive.

Problem 7 (AMO 2014) Initially, some 1×1 tiles in an $n \times n$ grid are coloured. Sally can colour any uncoloured tile if the tile shares an edge with at least three already coloured tiles. She discovers that by repeating this process, she can eventually colour all the tiles in the grid. Prove that the initial number of coloured tiles must be greater than $\frac{n^2}{3}$.

Problem 8 (CGMO 2019) For any sequence of numbers, an allowed operation consists of choosing three adjacent terms, say a, b, c, and replacing them with b, c, a, while keeping all other terms unchanged. Determine all integers $n \geq 3$ such that the sequence $1, 2, \ldots, n$ can be transformed into the reversed sequence $n, n-1, \ldots, 1$ in a finite number of operations.

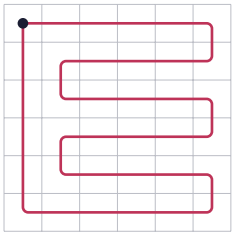
Problem 9 There are 2018 stones placed at various integer coordinates (x, y) on a Cartesian plane.

© 2026 Kevin Hu · kevinvariant08.github.io · CC BY-NC-ND 4.0

An allowed operation consists of choosing a grid point (a, b) that contains 4 or more stones, removing exactly 4 stones from this point, and placing one stone on each of its four adjacent grid points $(a, b-1)$, $(a, b+1)$, $(a-1, b)$ and $(a+1, b)$. Prove that after a finite number of operations, the number of stones at every grid point will be at most 3. Furthermore, prove that the final configuration of the stones is independent of the order in which the operations are performed.

3 Solutions

Problem 1 (MAT 2009)



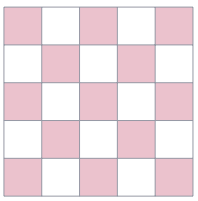
(i) For any even n, a tour can be constructed with a comb pattern. The path moves right across the entire top row, then weaves back and forth along the remaining rows, staying out of the left-most column, then moves left across the bottom row and returns straight up the left-most column to the start. The figure shows the case $n = 6$.

(ii) Yes. A tour forms a closed loop (a *Hamiltonian cycle*). Because the tour is a closed loop, the sequence of moves has no distinguished beginning or end. Shifting the start/finish point only permutes the same sequence of valid moves cyclically.

(iii) The robot visits every square of the $n \times n$ grid exactly once and returns to the start, which takes exactly one forward move per square. Thus $f = n^2$.

Note that the robot initially faces east and ends facing north. If the robot turns clockwise once more after the tour ends, the robot will have completed a whole number of full turns, so the number of clockwise right-angle turns is a multiple of 4. Hence $r + 1 \equiv 0 \pmod 4$.

(iv) Yes. The number of forward moves is still $f = n^2$, since the robot still visits each of the n^2 squares once and returns to the start. The turning argument of part (iii) also still applies: from leaving the start to arriving back, the robot turns through a whole number of full turns in total, whichever square the tour starts from. The form of the congruence depends on the start square. If the tour turns right at the start square, as the top-left square must, the robot arrives one right turn short of its starting direction and $r + 1 \equiv 0 \pmod 4$ exactly as before. If the tour turns left there, the robot is three right turns short and $r + 3 \equiv 0 \pmod 4$. If the start square lies on a straight section, the robot arrives facing the way it set off and $r \equiv 0 \pmod 4$.



(v) Colour the $n \times n$ grid in a chessboard pattern. Each move F takes the robot from a square of one colour to a square of the other colour. Any closed tour must therefore contain an even number of steps to return to the starting colour. However, a valid tour visits exactly n^2 squares and needs exactly n^2 steps. If n is odd, n^2 is odd, which contradicts the parity. Thus a tour is impossible. ■

Problem 2 (MAT 2020 extension)

Let M be the total number of sweets Miriam eats and A the total number of sweets Adam eats. We will prove that $A - M = 0$ for every possible arrangement, by finding an invariant under

swaps of adjacent days.

Consider an arbitrary sequence of weather, and suppose we swap a rainy day on day k with a sunny day on day $k+1$. We look at how this swap affects both totals.

The rainy day moves from day k to day $k+1$. Therefore Adam eats $k+1$ sweets instead of k for this rainy day, and A increases by exactly 1.

The sunny day now occurs one day earlier, on day k instead of day $k+1$, so Miriam eats exactly 1 extra sweet on day k. On days before k the number of sunny days so far is unchanged, and on days after k the number of sunny days so far is also unchanged. So M increases by exactly 1.

Since A and M both increase by 1, the difference $A-M$ is invariant under any swap of an adjacent (rain, sun) pair to (sun, rain). Any arrangement of the weather can be reached by adjacent swaps (as in bubble sort), so $A-M$ takes the same value for all possible sequences.

Hence we can evaluate $A-M$ for the simplest arrangement, in which all n rainy days come first, followed by all n sunny days. Adam eats sweets on days 1 to n, so his total is the sum of the first n integers,

$$A = 1+2+3+\cdots+n = \frac{n(n+1)}{2}.$$

Miriam eats 0 sweets on each of the first n days, as there has been no sun. On day $n+1$ she eats 1, on day $n+2$ she eats 2, and so on until day $2n$, when she eats n. Her total is

$$M = 1+2+3+\cdots+n = \frac{n(n+1)}{2}.$$

In this case $A-M=0$. Because $A-M$ is the same for all arrangements of the weather, $A-M=0$ always, so Miriam and Adam always eat the same number of sweets. ■

Problem 3 (BMO1 2024)

Rhian has a winning strategy. We use two state functions, a monovariant to prove that the game ends and an invariant to decide who wins.

First, the monovariant. Let n be the current number. A valid move gives $n' = |a-b|$, where $a \neq b$ and $ab = n$. Assuming without loss of generality that $a > b \geq 1$, we have $a - b < ab = n$. Therefore the number on the board decreases with every move. Furthermore, because the rules require $a \neq b$, the number 0 can never appear. The only positive integer with no factorisation into two distinct positive integers is 1. Thus the game is finite and always ends at the terminal state $n = 1$.

Now consider the parity of the number handed to a player. If a player receives an odd number n, then in any factorisation $n = ab$ both a and b are odd. The difference of two odd integers is even. Thus a player handed an odd number *must* hand back an even number. If a player receives an even number n, the player can always choose the factorisation $a = n$, $b = 1$. The resulting number is $n-1$, which is odd since n is even. Thus a player handed an even number can *always* choose to hand back an odd number.

Rhian starts the game with $n = 10^6$, which is even. Her winning strategy is to always hand an odd number to Jack, for example by always choosing the factorisation $a = n$, $b = 1$. Jack, receiving an odd number, is forced to hand an even number back to Rhian.

Because the number decreases, the game must eventually reach the terminal state $n = 1$. Since 1 is odd, and Jack is the *only* player who ever receives odd numbers, Jack will eventually be handed the 1, and so Jack loses. ■

© 2026 Kevin Hu · kevinvariant08.github.io · CC BY-NC-ND 4.0

Problem 4

If we let the newly written number be $z = x + y + xy$, then

$$z + 1 = x + y + xy + 1 = (x + 1)(y + 1).$$

This gives an invariant. If we take the quantity $v + 1$ for every number v on the board, the product of all these quantities stays the same after every operation. Let P be this product for the numbers initially on the board,

$$P = (1+1)\left(1+\frac{1}{2}\right)\left(1+\frac{1}{3}\right)\cdots\left(1+\frac{1}{2018}\right) = \frac{2}{1}\cdot\frac{3}{2}\cdot\frac{4}{3}\cdots\frac{2019}{2018} = 2019.$$

Let the final number on the board be F. Because the product is preserved throughout,

$$F + 1 = P \implies F + 1 = 2019 \implies F = 2018.$$

Since F is determined by the initial board and the invariant alone, the final value is 2018, independent of the order in which the pairs were chosen. ■

Problem 5 (RMM 2021)

Yes, for every $n \geq 2$ the board can be reduced to exactly two numbers.

We will prove that the set $\{1, 2, \ldots, n\}$ can be reduced to a set of even numbers $\{2, 4, 6, \ldots, 2m\}$. Because the operations $a + b$ and $|a - b|$ commute with multiplication by a constant, playing the game on $\{2, 4, \ldots, 2m\}$ is isomorphic to playing on $\{1, 2, \ldots, m\}$. So we can keep halving the size of the board.

Case 1, $n = 4k$ or $n = 4k - 1$. The odd numbers initially on the board are $1, 3, \ldots, 4k - 1$, and there are exactly $2k$ of them. We pair them from the outside in,

$$(1, 4k - 1),\ (3, 4k - 3),\ \ldots,\ (2k - 1, 2k + 1).$$

For every pair the sum is $4k$. The differences are $4k - 2, 4k - 6, \ldots, 2$. Note that $4k$ and all the differences are even numbers at most $4k$. Thus they are either already on the board, or equal to $4k$, which is added if $n = 4k - 1$. After erasing all the odd numbers through these pairs, the board contains exactly the even numbers $\{2, 4, 6, \ldots, 4k\}$. Dividing by 2, this reduces to the case $n = 2k$.

Case 2, $n = 4k + 1$ or $n = 4k + 2$, with $k \geq 2$. First we apply the operation to the pair $(2k + 1, 4)$. The sum is $2k + 5$ and the difference is $2k - 3$. Because $k \geq 2$, we have $2k - 3 \geq 1$ and $2k + 5 \leq 4k + 1$, so both results are odd numbers already on the board. Thus erasing $(2k + 1, 4)$ adds no new numbers. The remaining odd numbers are $1, 3, \ldots, 2k - 1$ and $2k + 3, \ldots, 4k + 1$. We pair them from the outside in,

$$(1, 4k + 1),\ (3, 4k - 1),\ \ldots,\ (2k - 1, 2k + 3).$$

For every pair the sum is $4k + 2$, and the differences are $4k, 4k - 4, \ldots, 4$. Note that the innermost pair gives the difference 4, which replaces the 4 erased in the first step. All the sums and differences are even numbers at most $4k + 2$. After these operations all odd numbers are erased, and the board contains exactly $\{2, 4, 6, \ldots, 4k + 2\}$. Dividing by 2, this reduces to the case $n = 2k + 1$.

Base cases. Case 2 requires $k \geq 2$, so we need to check $n = 5$ and $n = 6$, which correspond to $k = 1$. For $n = 5$ the board is $\{1, 2, 3, 4, 5\}$. Erasing $(2, 3)$ gives the sum 5 and the difference 1, both already present, so the board becomes $\{1, 4, 5\}$. Erasing $(1, 5)$ gives the sum 6 and the difference 4, and 4 is present. The board becomes $\{4, 6\}$, exactly two numbers. For $n = 6$ the board is $\{1, 2, 3, 4, 5, 6\}$. Erasing $(2, 3)$ gives 5 and 1, both present, so the board becomes $\{1, 4, 5, 6\}$. Erasing $(1, 5)$ gives 6 and 4, both present. The board becomes $\{4, 6\}$, exactly two numbers.

By strong induction, the board can always be reduced to two numbers. ■

© 2026 Kevin Hu · kevinvariant08.github.io · CC BY-NC-ND 4.0

Problem 5, alternative solution

Yes, this is always possible. We will prove a stronger lemma. *Any set of* 3 *positive integers* $\{a, b, c\}$ *can be reduced to exactly* 2 *numbers.* By repeatedly applying this lemma to any three numbers on the board, a set of n numbers will eventually reduce to exactly two. We prove the lemma by strong induction on $S = a + b + c$.

First, we define a macro-operation $(*)$ that doubles any two numbers and leaves the third unchanged,

$$\{a, b, c\} \xrightarrow{\text{operate on } a,b} \{a + b, |a - b|, c\} \xrightarrow{\text{operate on } a+b,|a-b|} \{2a, 2b, c\}.$$

Because the game operations are linear, if all numbers on the board share a common divisor d, the game is isomorphic to playing with the numbers divided by d.

Case 1, $\gcd(a, b, c) = d > 1$. We divide by d, and the state reduces to $\{a/d, b/d, c/d\}$. The new sum is $S/d < S$, so by strong induction this reduces to 2 numbers.

Case 2.1, $\gcd(a, b, c) = 1$ *and at least one number is even.* Without loss of generality c is even. Apply $(*)$ to a and b to get $\{2a, 2b, c\}$. Now all three numbers are even. Dividing by 2, the state becomes $\{a, b, c/2\}$. The new sum is $a + b + c/2 < S$, so by induction this reduces to 2 numbers.

Case 2.2, $\gcd(a, b, c) = 1$ *and all three numbers are odd.* Without loss of generality $a > b$. First operate on a and b, and the board becomes $\{a + b, a - b, c\}$, where $a + b$ and $a - b$ are even and c is still odd. Then apply $(*)$ to $a - b$ and c, and the board becomes $\{a + b, 2(a - b), 2c\}$. Finally apply $(*)$ to $a + b$ and $2c$, and the board becomes $\{2(a + b), 2(a - b), 4c\}$.

Since a and b are odd, both $a + b$ and $a - b$ are even, so $2(a + b)$ and $2(a - b)$ are multiples of 4. Now all three numbers on the board are multiples of 4. Dividing by 4, the state becomes

$$\left\{\frac{a+b}{2}, \frac{a-b}{2}, c\right\}.$$

The new sum is $\frac{a+b}{2} + \frac{a-b}{2} + c = a + c$. Because $b > 0$, we have $a + c < a + b + c = S$, so by strong induction this reduces to 2 numbers.

Thus in every case the triple reduces to two numbers, and any $n \geq 2$ starting numbers can be reduced to exactly two. ■

Problem 6 (Russia 2022)

We define the state function $f(x, y) = y^2 - 4x$ and look at how f behaves under the two allowed operations. For the first operation,

$$\begin{aligned} f(x + y, y + 1) &= (y + 1)^2 - 4(x + y) \\ &= y^2 + 2y + 1 - 4x - 4y \\ &= (y - 1)^2 - 4x \\ &= f(x, y - 1). \end{aligned}$$

Thus the value of f is invariant under the first operation. For the second operation,

$$f(x, xy) = (xy)^2 - 4x = x^2y^2 - 4x, \qquad f\left(\frac{1}{x}, y\right) = y^2 - \frac{4}{x}.$$

Note that $f(x, xy) = x^2 \cdot f\left(\frac{1}{x}, y\right)$. Because $x^2 > 0$ (as $x \neq 0$), multiplying by x^2 does not change the sign. Thus the *sign* of f is invariant under the second operation.

The board starts with the pair $(1,1)$, and $f(1,1) = 1^2 - 4 \cdot 1 = -3 < 0$. Since the operations preserve either the exact value or the sign of f, every pair (x,y) written on the board satisfies $f(x,y) < 0$. Therefore, for every pair on the board,

$$y^2 - 4x < 0 \implies x > \frac{y^2}{4} \geq 0.$$

So $x > 0$, and the first number in every pair is positive. ■

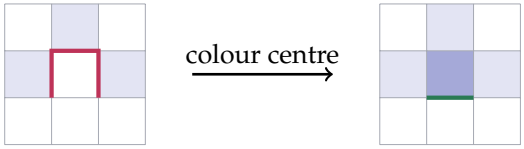
Problem 7 (AMO 2014)

We use a geometric monovariant. To avoid special cases at the border of the grid, we place the $n \times n$ grid inside a larger $(n+2) \times (n+2)$ grid. The outer ring of this new grid consists of uncoloured tiles, which can never be coloured.

Let S be the total number of edges shared by a coloured tile and an uncoloured tile, that is, the perimeter of the coloured region. Let k be the initial number of coloured tiles. Since each coloured tile has exactly 4 edges, the initial perimeter is largest when all k tiles are isolated. Thus

$$S_{\text{initial}} \leq 4k.$$

When Sally colours a new tile, the tile must have at least 3 coloured neighbours. If the tile has exactly 3 coloured neighbours, colouring the tile removes 3 of these shared edges and creates 1 new one, so S changes by -2. If the tile has 4 coloured neighbours, colouring the tile removes 4 such edges and creates none, so S changes by -4. In every case each operation decreases S by *at least* 2.



Before: 3 shared edges After: 1 new edge (net -2)

The process stops when all n^2 tiles of the $n \times n$ grid are coloured. At this point the coloured region is a solid $n \times n$ square surrounded by the uncoloured ring, and the final perimeter is the boundary of this square,

$$S_{\text{final}} = 4n.$$

Since we started with k coloured tiles and ended with n^2 coloured tiles, exactly $n^2 - k$ operations were performed. Because each operation decreases S by at least 2,

$$S_{\text{initial}} - S_{\text{final}} \geq 2(n^2 - k).$$

Substituting $S_{\text{initial}} \leq 4k$ and $S_{\text{final}} = 4n$ gives

$$4k - 4n \geq 2n^2 - 2k, \qquad \text{so} \qquad k \geq \frac{n^2 + 2n}{3}.$$

Since $n \geq 1$, we have $\frac{2n}{3} > 0$. Therefore

$$k \geq \frac{n^2}{3} + \frac{2n}{3} > \frac{n^2}{3},$$

and more than $\frac{n^2}{3}$ tiles must be coloured initially. ■

Problem 8 (CGMO 2019)

The sequence can be reversed if and only if $n \equiv 0 \pmod 4$ or $n \equiv 1 \pmod 4$.

The operation $(a, b, c) \to (b, c, a)$ is a cyclic shift of 3 elements. This is the same as two adjacent transpositions (swap a, b, then swap a, c). Because the number of swaps is even, the operation preserves the parity of the total number of inversions in the sequence. The initial sequence $1, 2, \ldots, n$ has 0 inversions, an even number. The target sequence $n, n-1, \ldots, 1$ has exactly $\frac{n(n-1)}{2}$ inversions. For the target to be reachable, $\frac{n(n-1)}{2}$ must be even, so $n(n-1)$ must be divisible by 4. This happens only when $n \equiv 0 \pmod 4$ or $n \equiv 1 \pmod 4$.

Conversely, note that applying the operation twice to the same three elements gives $(a, b, c) \to (b, c, a) \to (c, a, b)$. This macro-operation moves c two positions to the left (or a two positions to the right) without changing the relative order of the other elements. By repeating the macro-operation, we can move any element left or right by any even number of positions without disturbing the rest of the sequence.

For $n = 4$, start with $(1, 2, 3, 4)$. Move 4 left by two positions to get $(1, 4, 2, 3)$. Move 1 right by two positions to get $(4, 2, 1, 3)$. Move 3 left by two positions to get $(4, 3, 2, 1)$. For $n = 5$, start with $(1, 2, 3, 4, 5)$. Move 5 left by four positions (two jumps of two) to get $(5, 1, 2, 3, 4)$. Then apply the $n = 4$ algorithm to the remaining block to get $(5, 4, 3, 2, 1)$.

We first treat $n \equiv 0 \pmod 4$ by induction, with $n = 4$ as the base case. Assume the reversal is possible for $n - 4$, and consider a sequence of length n. We split the sequence into two blocks, $A = (1, \ldots, n-4)$ and $B = (n-3, n-2, n-1, n)$. Since $n - 4$ is a multiple of 4, block A has even length, so we can use the macro-operation to move every element of block B to the left of block A, jumping 2 elements at a time. This moves block B to the front without changing the internal order of A or B. The sequence is now $(n-3, n-2, n-1, n, 1, 2, \ldots, n-4)$. Finally, we apply the $n = 4$ algorithm to reverse the first block and the induction hypothesis to reverse the second block, which gives $(n, n-1, \ldots, 1)$.

Now let $n \equiv 1 \pmod 4$. As in the case $n = 5$, move n to the front, which is $n - 1$ places and so an even number of places. The sequence is now $(n, 1, 2, \ldots, n-1)$, and the block $(1, \ldots, n-1)$ has length $n - 1 \equiv 0 \pmod 4$, so the first case reverses the block. This gives $(n, n-1, \ldots, 1)$. Thus all $n \equiv 0, 1 \pmod 4$ are achievable. ■

Problem 9

Termination. Let $n = 2018$ and denote the coordinates of the stones by (x_k, y_k). Define two state functions,

$$f = \sum_{k=1}^{n} (x_k^2 + y_k^2), \qquad g = \sum_{i<j} \big(|x_i - x_j| + |y_i - y_j|\big).$$

An operation at (a, b) moves 4 stones to $(a \pm 1, b)$ and $(a, b \pm 1)$. Expanding the sum of squares gives $\Delta f = 4$, so after K operations $f_K = f_0 + 4K$.

To find Δg, we split the pairwise distances into two kinds. The Manhattan distances among the 4 moving stones increase in total from 0 to 12 (two opposite pairs at distance 2 and four adjacent pairs at distance 2). For any stationary stone at (x, y), the total distance to the 4 moving stones changes from $4(|x - a| + |y - b|)$ to

$$\big(|x - a - 1| + |x - a + 1| + 2|x - a|\big) + \big(|y - b - 1| + |y - b + 1| + 2|y - b|\big).$$

By the triangle inequality, $|u - 1| + |u + 1| \geq 2|u|$, so this new distance is at least $4|x - a| + 4|y - b|$. Thus the cross distances cannot decrease. Therefore $\Delta g \geq 12$, which gives $g_K \geq g_0 + 12K$.

© 2026 Kevin Hu · kevinvariant08.github.io · CC BY-NC-ND 4.0

We relate g to f by the Cauchy-Schwarz inequality. For the x-coordinates,

$$\begin{aligned}\Big(\sum_{i<j}|x_i-x_j|\Big)^2 &\le \frac{n(n-1)}{2}\sum_{i<j}(x_i-x_j)^2 = \frac{n(n-1)}{2}\left[n\sum_{i=1}^{n}x_i^2-\Big(\sum_{i=1}^{n}x_i\Big)^2\right]\\ &\le \frac{n^2(n-1)}{2}\sum_{i=1}^{n}x_i^2.\end{aligned}$$

The same bound for the y-coordinates gives $g_K^2 \le C\cdot f_K$, where C is a constant depending only on n. Substituting the bounds above gives $(g_0+12K)^2 \le C(f_0+4K)$. A quadratic function of K cannot be bounded by a linear function for all K, so the total number of operations K is finite.

Uniqueness. Let Φ be the initial state, and let T_k denote the operation of firing the grid point k. Operations commute, $T_aT_b(\Phi) = T_bT_a(\Phi)$, because firing a only adds stones to the neighbours of a, which never invalidates a pending valid move at b.

Suppose two maximal finite sequences of operations lead to termination. If the first sequence begins with T_{a_1}, then a_1 must be fired at some point in the second sequence. Let T_{b_i} be the first occurrence of firing a_1 in the second sequence. Because operations commute, we can slide T_{b_i} to the very front of the second sequence without changing the final state,

$$T_{b_m}\cdots T_{b_i}\cdots T_{b_1}(\Phi) = T_{b_m}\cdots T_{a_1}\cdots T_{b_1}(\Phi) = T_{b_m}\cdots T_{b_1}T_{a_1}(\Phi).$$

We can now cancel T_{a_1} from the front of both sequences. By repeatedly sliding and cancelling every later operation, we find that both sequences consist of the same multiset of operations, so the final arrangement is unique. ■

© 2026 Kevin Hu · kevinvariant08.github.io · CC BY-NC-ND 4.0